STAT 305 HW 1 1/18/26.

1.

$$p(x) = \begin{cases} \frac{x}{10} & x = 1, 2, 3, 4 \\ 0, & \text{otherwise} \end{cases}$$

A. $p(x=1) = 1/10$, $p(x=2) = 2/10$, $p(x=3) = 3/10$, $p(x=4) = 4/10$

B. $E(x) = \sum x p(x) = \frac{1}{10} + 2\left(\frac{2}{10}\right) + 3\left(\frac{3}{10}\right) + 4\left(\frac{4}{10}\right)$
$= 30/10 = 3$

C. $V(x) = E(x^2) - E(x)^2 = \left(\frac{1}{10} + \cancel{4}\left(\frac{2}{10}\right) + \cancel{9}\left(\frac{3}{10}\right) + \cancel{16}\left(\frac{4}{10}\right)\right)$ 8 27 64
$- 3^2 = 10 - 9 = 1$ .5 3.5 = 1

D. $X_1, X_2 \underset{iid}{\sim} p(x)$.

$E(4X_1 - 3X_2) = 4E(X_1) - 3E(X_2) = 4(3) - 3(3) = 3$

E. $Var(4X_1 - 3X_2) = 16(1) + 9(1) = 25$

In general we have $Var(aX_1 + bX_2) = E\left((aX_1 + bX_2) - (a\mu_1 + b\mu_2)\right)^2$ $a, b \in \mathbb{R}$

$= E\left((aX_1 - a\mu_1) + (bX_1 - b\mu_2)\right)^2$

$= E\left((aX_1 - a\mu_1)^2 + (bX_1 - b\mu_2)^2 + 2(aX_1 - a\mu_1)(bX_1 - b\mu_2)\right)$

$var(aX_1) + var(bX_2) + 2cov(aX_1, bX_2)$.

Since $X_1, X_2$ are ~~thats~~ iid.
$cov(aX_1, bX_2) = 0$ then

$= a^2 var(X_1) + b^2 var(X_2)$

2. $f(y) = \begin{cases} \frac{0.72135}{y} & 1 \le y \le 4 \\ 0, & \text{otherwise} \end{cases}$

A. $f(2) = \frac{0.72135}{2} = 0.360675$

$P(1 \le y \le 2) = \int_1^2 \frac{0.72135}{y} dy \approx 0.5$

B. $E(Y) = \int_1^4 y\left(\frac{0.72135}{y}\right) dy = 0.72135(3)$. $0.72135 \ln|y| \Big|_1^2 = \ln(2) \cdot 0.72135$

$\approx 2.16405$

C. $var(Y) = \int_1^4 y^2\left(\frac{0.72135}{y}\right) dy - (2.164)^2 \approx 0.7270$

D. $E(2Y+1) = 2E(x) + 1 = 2(2.164) + 1 \approx 5.3281$

E. $var(2Y+1)$
$= 4 var(Y)$
$= 4(0.7270)$
$\approx 2.90805$